Student Name: ..... Enrollment Number: .....

Mid-Term Examination- November 2022

Program: B.Tech **Semester: 3rd**

Paper Code: ECC-213 **Paper Name: Electromagnetic Field Theory**

Time: 90 Minutes **Maximum Marks: 30**

[Question 1 is compulsory. Attempt 2 more.]

Q1. (a) If your friend told you that the permittivity of a non-dispersive material is not a constant but a delta function, would you agree with him/ her? Give reason. (2)

(b) Using Maxwell's equations, derive the relationship between conductivity and effective permittivity in 2- 3 steps. (2)

(c.) Though scalar potential V(ρ,φ, z, t) has a z- dependent term, why is an electric field in the z-direction still zero? (2)

(d) Write down expressions for increment surfaces in (i) the Cylindrical coordinate system & (ii) the Spherical coordinate system. (2)

(e) Derive differential form of Faraday's Law of EMI. (2)

Q2. (a) Find the distance between (2, π/ 6, 0) and (1, π, 2). (2)

(b) Transform the vector $\bar{A}$, given in spherical coordinates as $\bar{A} = 5r\hat{i}_r + 2\sin\phi\hat{i}_\theta + 2\cos(\text{theta})\hat{i}_{(\text{phi})}$ into the cartesian coordinate system. (6)

(c) Evaluate the magnitude of vector $\bar{A}$ in (b) above at the point P given by P(3,4,12). (2)

Q3. (a) If $F = x^3\hat{i}_x + x^2yz\hat{i}_y + x^2z\hat{i}_z$, verify the divergence theorem when the region is bounded by the curve $x^2 + y^2 = 16$ and the plane at z=0 & z=2. (5)

(b) Derive $(\tan\theta_1/\tan\theta_2) = (\varepsilon_1/\varepsilon_2)$ for a boundary separated by two dielectric mediums with permittivities ε_1 and ε_2 where θ_1 and θ_2 are angles of incidence and refraction respectively. (5)

Q4. (a) Using Biot Savart's Law or otherwise, find the magnetic field at the center of a square loop where 'a' is the distance from its center to the side and the loop carries a steady current, I. (5)

(b) Derive the differential form of modified Ampere's Circuital Law from its integral form using Stoke's Theorem. (5)